

DH2642 Project Proposal: GitHub Hero Quest

1. Group Members

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2. Name of the App

[GitHub Hero Quest]

3. Short Description

A gamified dashboard that transforms GitHub repository activity into a team-based “Quest.” The app tracks commits and Pull Requests in real-time, converting them into Experience Points (XP) and levels. Teams can set collective goals to unlock achievements and monitor a live Hero Leaderboard.

4. Frameworks to be used

- **Frontend:** React
- **State Management:** React Context API
- **Backend/Persistence:** Firebase
- **Styling:** Tailwind CSS

5. API to be used

- **GitHub REST API:** To fetch repository statistics, commit history, and PR statuses.

6. Data Classification

- **Data from API (External):**
 - User profiles (avatars, usernames).
 - Repository activity (commit timestamps, frequencies).

- Pull Request metadata (status: open/merged/closed).
- **App Specific Data (Firebase):**
 - **Game Logic:** XP scoring rules and leveling thresholds.
 - **User Progress:** Current levels, earned badges, and total XP.
 - **Quests:** Custom team objectives, target values, and deadlines.
 - **Leaderboard:** Real-time ranking and historical snapshots.

7. Quick Mock-up

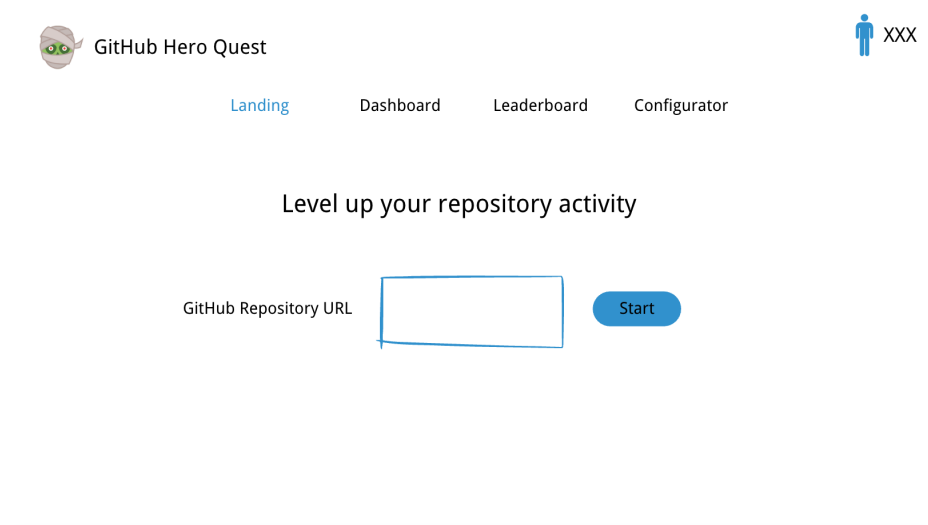


Figure 1: landing Page

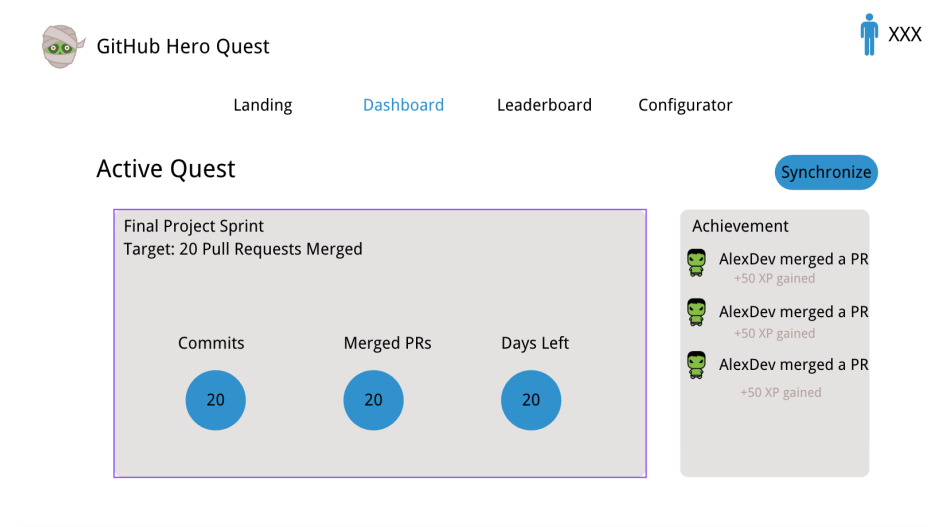


Figure 2: Main Quest Dashboard



Hero Hall

Current rankings based on XP earned this semester

Name	Class / Title	Current Level	Total XP
	level 1		
	level 2		
	level 3		

Figure 3: Leaderboard



Quest Name

Target

Deadline

[Submit](#)

Figure 4: Quest Configurator